

Coordination diagrams: single setting vs dual setting on R2

IEEE 13-node feeder, 4.05 MVA DG at node 692 - DSDR recloser-fuse coordination study

What is compared. The existing scheme has one setting on the mid-line recloser R2: it is non-directional and uses its forward setting for every fault, including the DG's reverse current for faults above it. As a dual-setting directional recloser (DSDR) R2 keeps the forward setting for faults below it and uses a second, reverse setting (eq. 12) for faults above it. **Both panels of every diagram use the same fuse sizes (the final design)**, so the only difference between left and right is R2's setting.

Device	Setting	Pickup / curve start	Time dial (fast / delayed)
R1 (GE IAC77, at RG60)	one setting, forward	720 A; curve from 1080 A	0.5 / 10
R2 forward (CDG34, at 671)	used by both schemes	plug 300 / 600 A; curve from 600 / 1200 A	0.1 / 1.0
R2 reverse (CDG34 curve)	dual setting only	plug 150 / 300 A; curve from 300 / 600 A (eq. 12: $1.25 \times 252.2 \text{ A} = 315 \text{ A}$)	0.1 / 1.0

Fault impedance. Each case states its fault impedance Z_f : 0 ohm (bolted) or the value used for that figure (1.0 ohm for Figs. 9 and 11, 1.5 ohm for Fig. 12). Currents are initial symmetrical short-circuit currents from PowerFactory (complete method), largest phase. Direction: + forward = away from the grid; for faults above R2 the current through R2 is the DG's contribution and flows in reverse (confirmed for all 48 bolted faults above R2 in the short-circuit database).

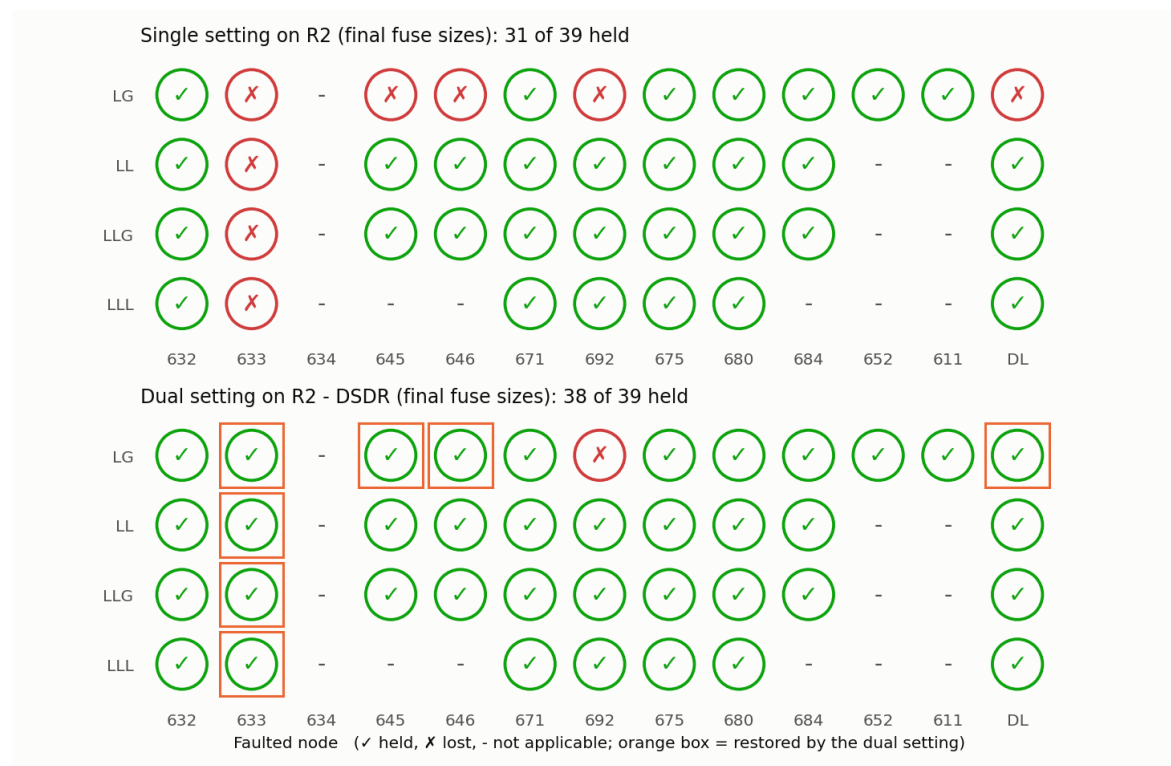
Fuse saving holds when every recloser that feeds the fault trips on its fast curve before the primary fuse melts, the fuse clears before the recloser's delayed trip, and the backup fuse keeps the 75 % margin of eq. (8). No breaker time or safety margin is added.

Fault table

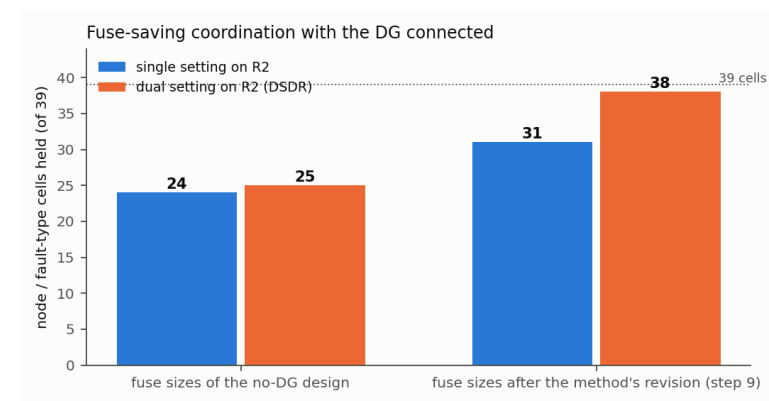
Case	Figure	Fault	Z_f (ohm)	Fault-path current	R1 current, direction	R2 current, direction	R2 fast: single -> dual	Fuse melts	Single setting	Dual setting	No-DG-design fuses: single / dual
01	Fig. 8	LG fault at 611 (LG C)	0	2236 A	1814 A, forward	1718 A, forward	0.128 s -> 0.128 s	0.271 s	held	held	held / held
02	Fig. 9	LLG fault at the middle of 692-675 (LLG ABC)	1	2084 A	1445 A, forward	1310 A, forward	0.204 s -> 0.204 s	0.618 s	held	held	held / held
03	Fig. 11	LL fault at 646 (LL BC)	1	2384 A	1899 A, forward	847 A, reverse	0.500 s -> 0.131 s	1.734 s	LOST	LOST	held / held
04	Fig. 12	LL fault at 645 (LL BC)	1.5	2137 A	1694 A, forward	783 A, reverse	0.605 s -> 0.149 s	9.129 s	LOST	LOST	held / held
05	Fig. 13	3-phase fault at 10 % of 632-633 (LLL ABC)	0	6398 A	4645 A, forward	1777 A, reverse	0.121 s -> 0.052 s	0.100 s	LOST	held	LOST / LOST
06	Figs. 15, 16	LL fault at 646 (LL BC)	0	3672 A	2785 A, forward	1114 A, reverse	0.278 s -> 0.086 s	0.415 s	held	held	LOST / LOST
07	Fig. 14 / 17 cell	LG fault at 646 (phase c) (LG C)	0	2732 A	2359 A, forward	469 A, reverse	no trip -> 0.396 s	1.057 s	LOST	held	LOST / LOST
08	Fig. 14 / 17 cell	LG fault at 633 (phase b) (LG B)	0	3443 A	2874 A, forward	644 A, reverse	0.974 s -> 0.210 s	0.502 s	LOST	held	LOST / LOST
09	Fig. 14 / 17 cell	LG fault at 692 (phase a) (LG A)	0	2100 A	2197 A, forward	2182 A, forward	0.089 s -> 0.089 s	2.842 s	LOST	LOST	held / held

Single / dual setting columns: final fuse sizes (as in the diagrams). Last column: the same faults with the fuse sizes of the no-DG design. Cases lost under both settings: 03 - series F646/F632: TCT $3.872 > 0.75 \times \text{MMT } 3.701$; 04 - R1 delayed (10.145 s) before F632 clears (25.861 s); 09 - R2 delayed (3.958 s) before F671-1 clears (7.580 s).

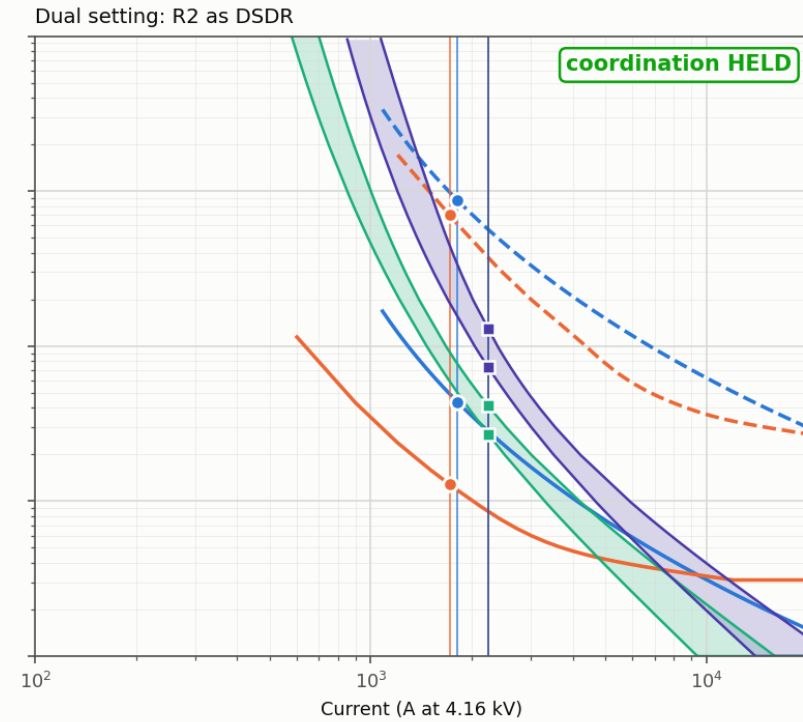
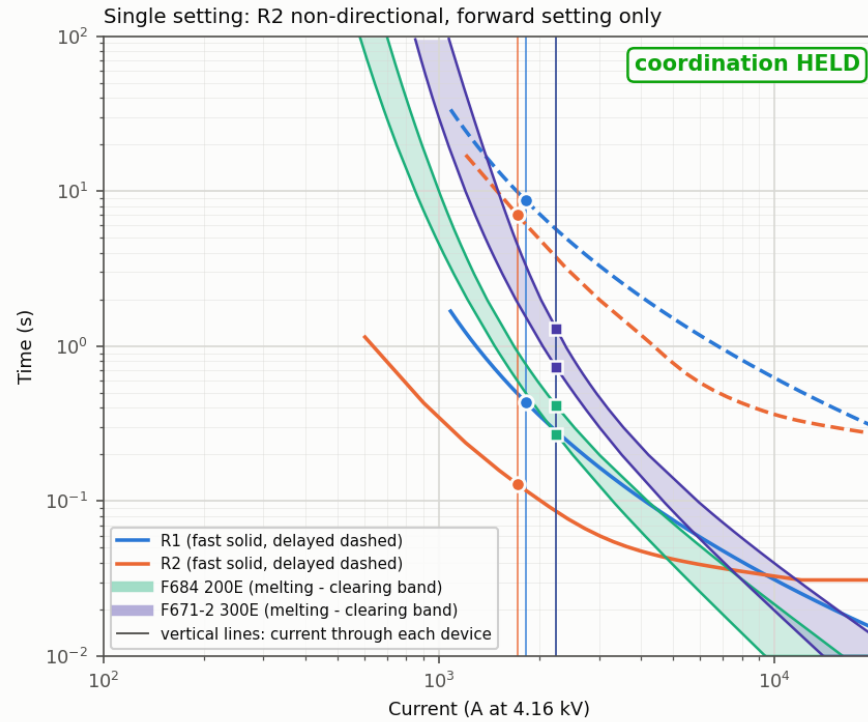
All 39 node / fault-type cells



Restored by the dual setting (final fuses): 633 LG/LL/LLG/LLL, 645 LG, 646 LG, DL LG. Still lost: 692 LG.

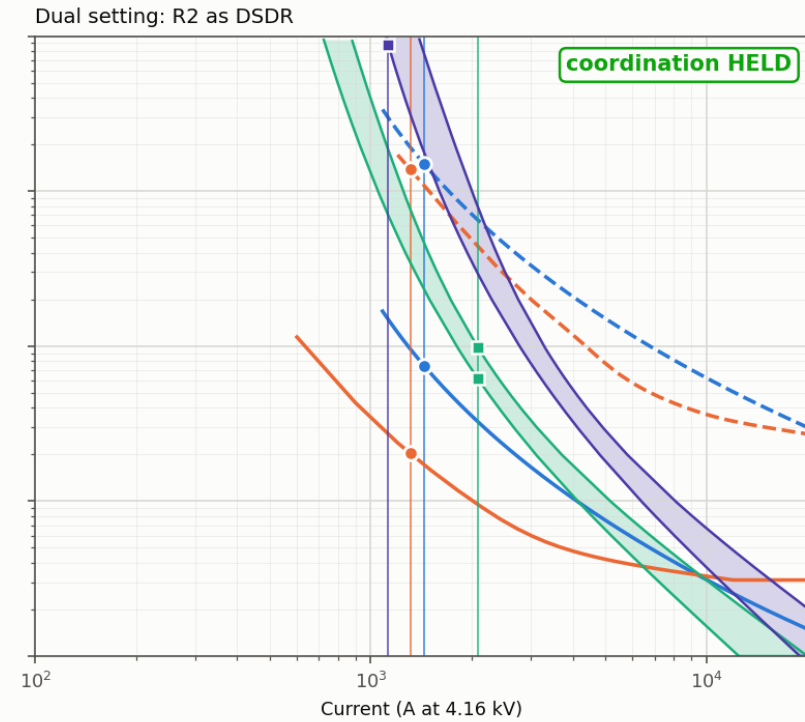
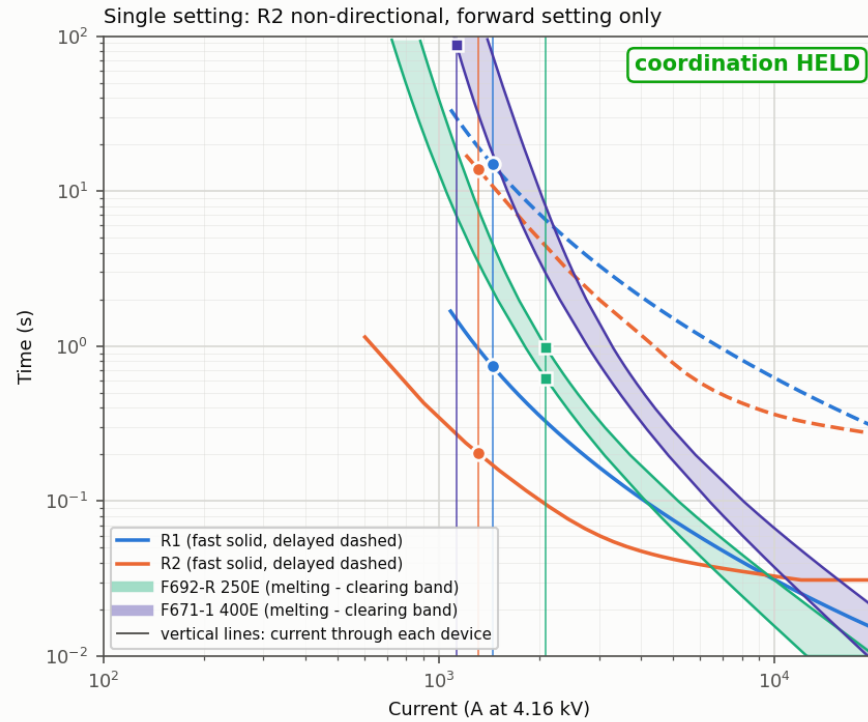


Case 01 - LG fault at 611, $Z_f = 0$ ohm (bolted), DG 4.05 MVA connected at 692 (Fig. 8)



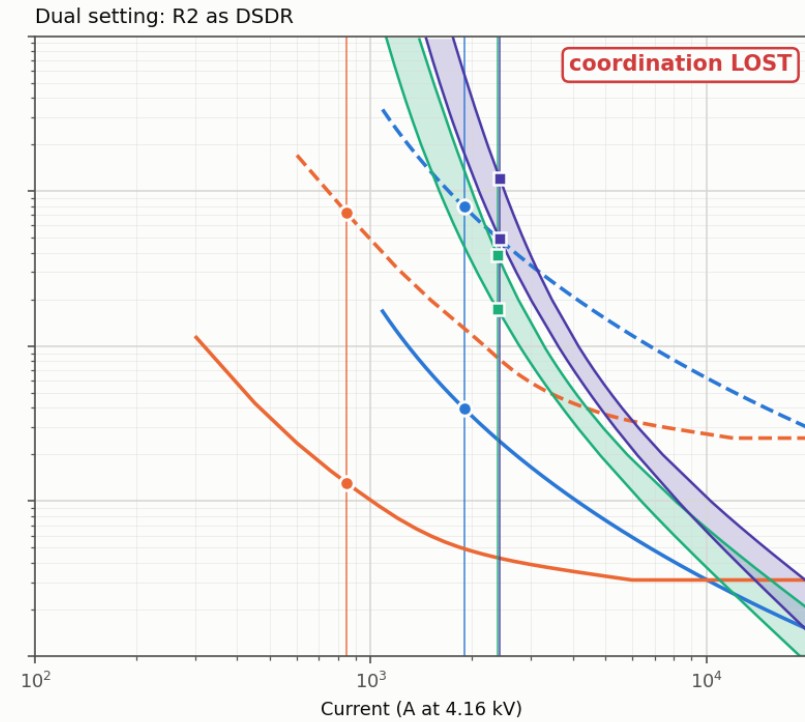
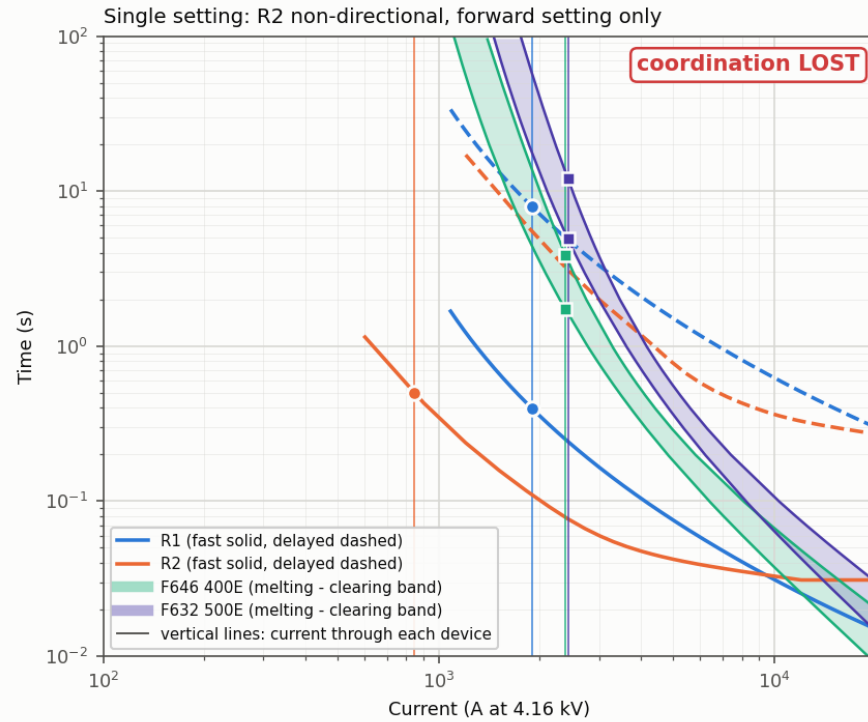
Quantity	Single setting	Dual setting (DSDR)
LG fault at 611	LG, phases C	(same fault and currents)
Fault impedance Z_f	0 ohm (bolted)	(same fault and currents)
Fault-path current (F684)	2236 A	(same fault and currents)
Current through R1, direction	1814 A, forward (RG60 -> 632), grid contribution	(same fault and currents)
Current through R2, direction	1718 A, forward (632 -> 671), grid contribution	(same fault and currents)
R2 setting applied (fast / delayed)	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0
R1 fast / delayed	0.435 s / 8.709 s	0.435 s / 8.709 s
R2 fast / delayed	0.128 s / 6.986 s	0.128 s / 6.986 s
Primary fuse melts / clears	F684 200E: 0.271 s / 0.414 s	F684 200E: 0.271 s / 0.414 s
Backup fuse melts	F671-2 300E: 0.732 s	F671-2 300E: 0.732 s
Fuse saving	HELD	HELD

Case 02 - LLG fault at the middle of 692-675, $Z_f = 1.0$ ohm, DG 4.05 MVA connected at 692 (Fig. 9)



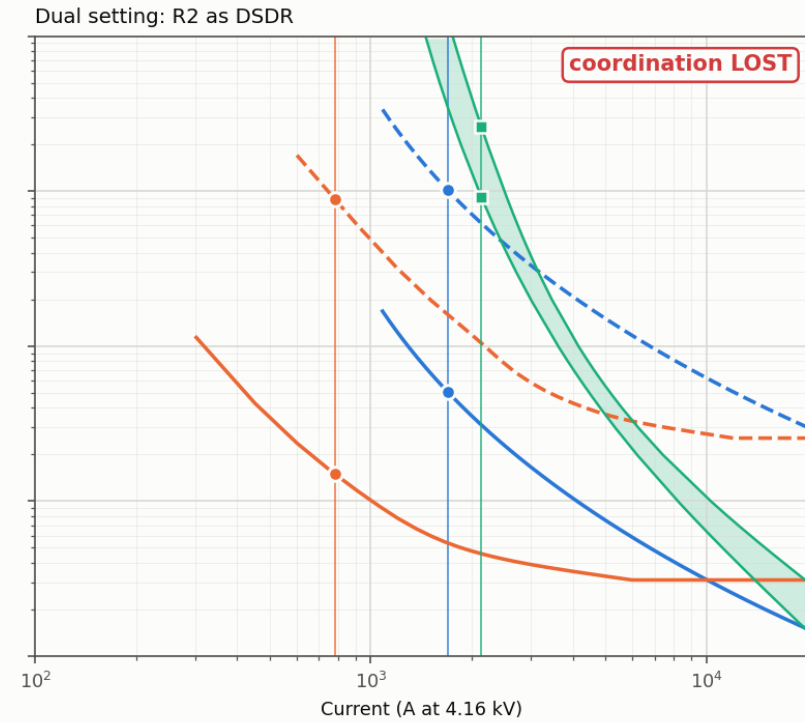
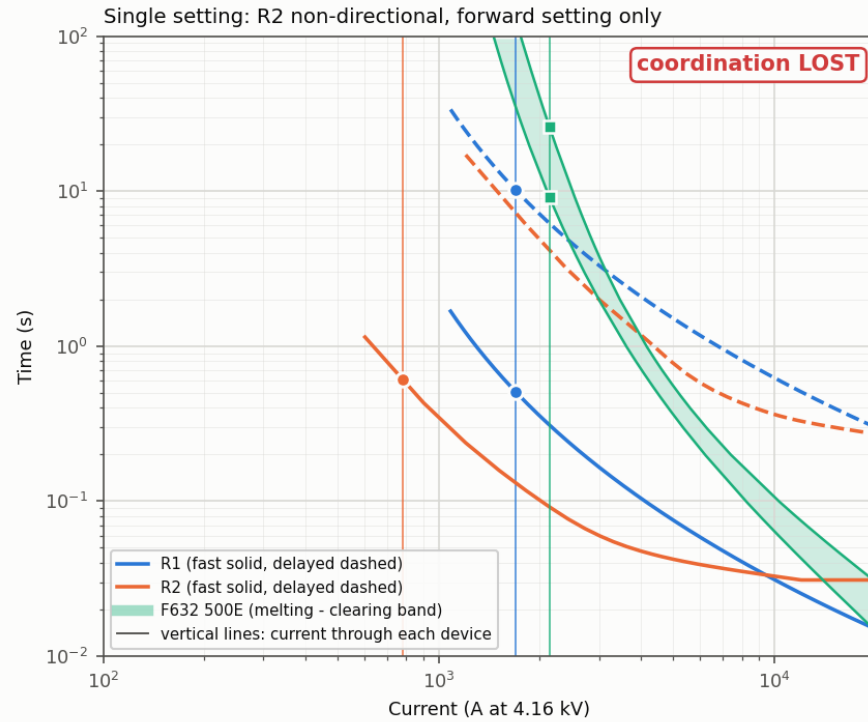
Quantity	Single setting	Dual setting (DSDR)
LLG fault at the middle of 692-675	LLG, phases a-b-c	(same fault and currents)
Fault impedance Z_f	1.0 ohm	(same fault and currents)
Fault-path current (F692-R)	2084 A	(same fault and currents)
Current through R1, direction	1445 A, forward (RG60 -> 632), grid contribution	(same fault and currents)
Current through R2, direction	1310 A, forward (632 -> 671), grid contribution	(same fault and currents)
R2 setting applied (fast / delayed)	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0
R1 fast / delayed	0.742 s / 14.843 s	0.742 s / 14.843 s
R2 fast / delayed	0.204 s / 13.798 s	0.204 s / 13.798 s
Primary fuse melts / clears	F692-R 250E: 0.618 s / 0.981 s	F692-R 250E: 0.618 s / 0.981 s
Backup fuse melts	F671-1 400E: 87.643 s	F671-1 400E: 87.643 s
Fuse saving	HELD	HELD

Case 03 - LL fault at 646, $Z_f = 1.0$ ohm, DG 4.05 MVA connected at 692 (Fig. 11)



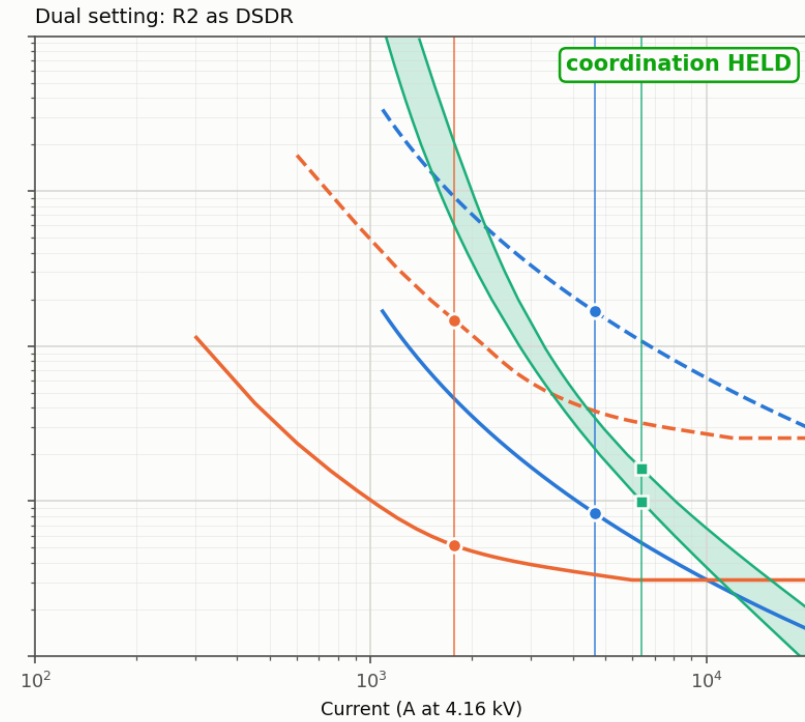
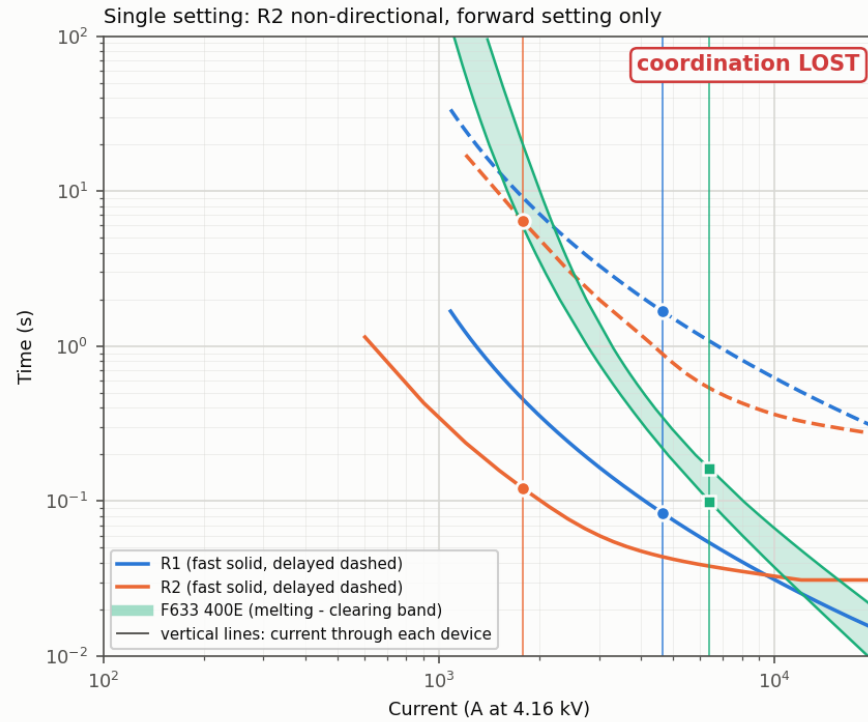
Quantity	Single setting	Dual setting (DSDR)
LL fault at 646	LL, phases BC	(same fault and currents)
Fault impedance Z_f	1.0 ohm	(same fault and currents)
Fault-path current (F646)	2384 A	(same fault and currents)
Current through R1, direction	1899 A, forward (RG60 -> 632), grid contribution	(same fault and currents)
Current through R2, direction	847 A, REVERSE (671 -> 632), DG contribution only	(same fault and currents)
R2 setting applied (fast / delayed)	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0	REVERSE: plug 150/300 A, starts 300/600 A, TMS 0.1/1.0
R1 fast / delayed	0.395 s / 7.895 s	0.395 s / 7.895 s
R2 fast / delayed	0.500 s / no trip	0.131 s / 7.246 s
Primary fuse melts / clears	F646 400E: 1.734 s / 3.872 s	F646 400E: 1.734 s / 3.872 s
Backup fuse melts	F632 500E: 4.935 s	F632 500E: 4.935 s
Fuse saving	LOST: series F646/F632: TCT 3.872 > 0.75 x MMT 3.701	LOST: series F646/F632: TCT 3.872 > 0.75 x MMT 3.701

Case 04 - LL fault at 645, $Z_f = 1.5 \text{ ohm}$, DG 4.05 MVA connected at 692 (Fig. 12)



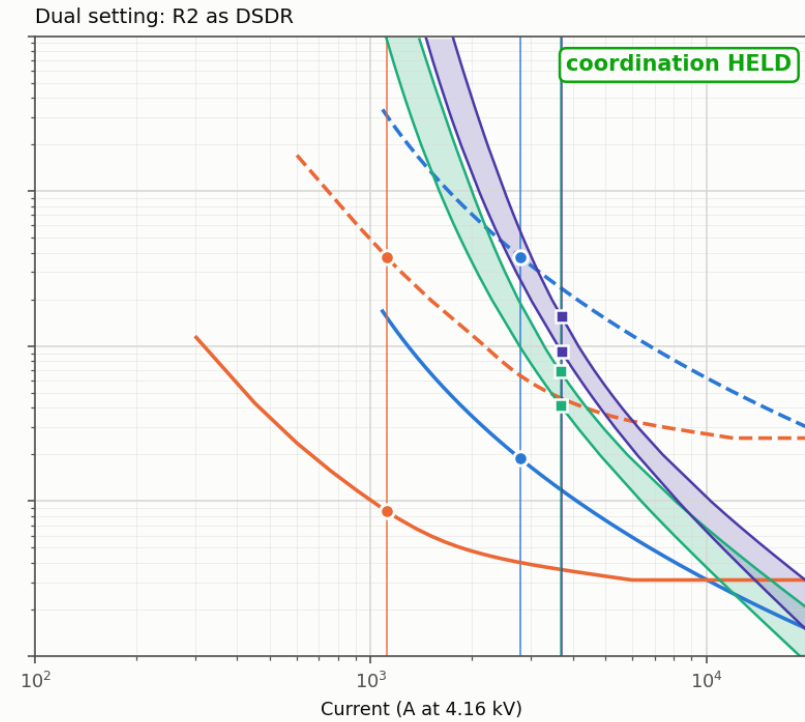
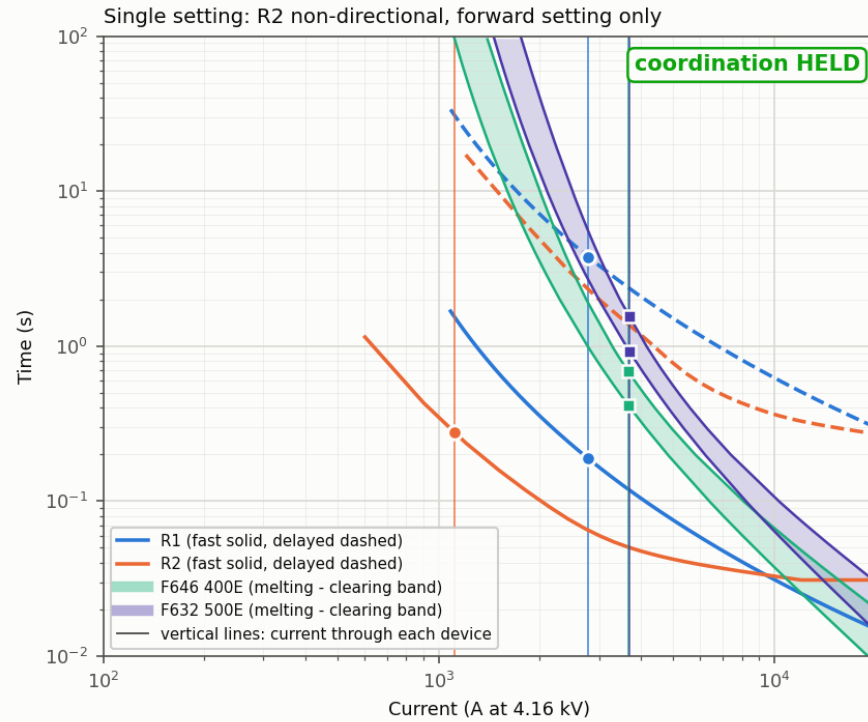
Quantity	Single setting	Dual setting (DSDR)
LL fault at 645	LL, phases BC	(same fault and currents)
Fault impedance Z_f	1.5 ohm	(same fault and currents)
Fault-path current (F632)	2137 A	(same fault and currents)
Current through R1, direction	1694 A, forward (RG60 -> 632), grid contribution	(same fault and currents)
Current through R2, direction	783 A, REVERSE (671 -> 632), DG contribution only	(same fault and currents)
R2 setting applied (fast / delayed)	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0	REVERSE: plug 150/300 A, starts 300/600 A, TMS 0.1/1.0
R1 fast / delayed	0.507 s / 10.145 s	0.507 s / 10.145 s
R2 fast / delayed	0.605 s / no trip	0.149 s / 8.816 s
Primary fuse melts / clears	F632 500E: 9.129 s / 25.861 s	F632 500E: 9.129 s / 25.861 s
Backup fuse melts	-	-
Fuse saving	LOST: R1 delayed (10.145 s) before F632 clears (25.861 s)	LOST: R1 delayed (10.145 s) before F632 clears (25.861 s)

Case 05 - 3-phase fault at 10 % of 632-633, $Z_f = 0$ ohm (bolted), DG 4.05 MVA connected at 692 (Fig. 13)



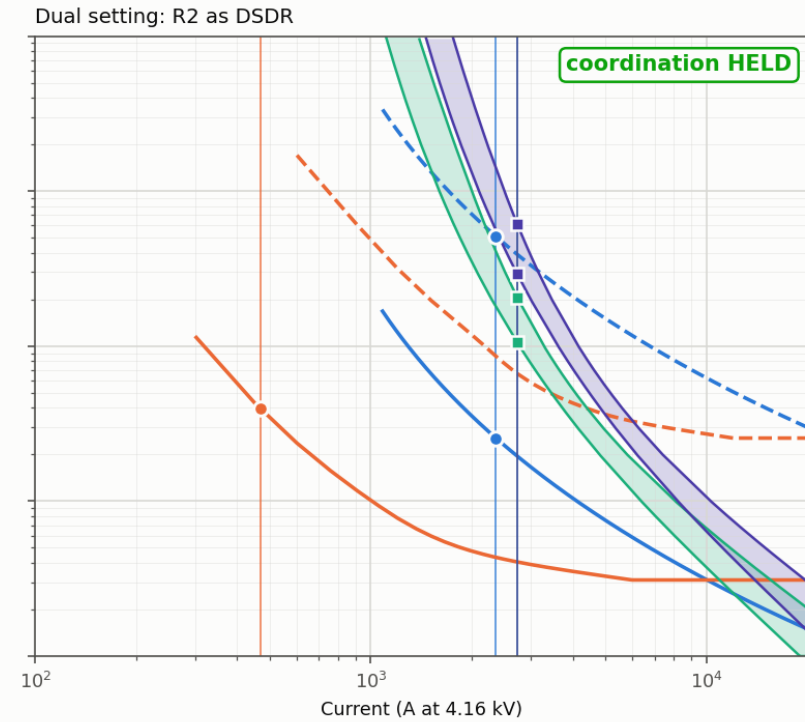
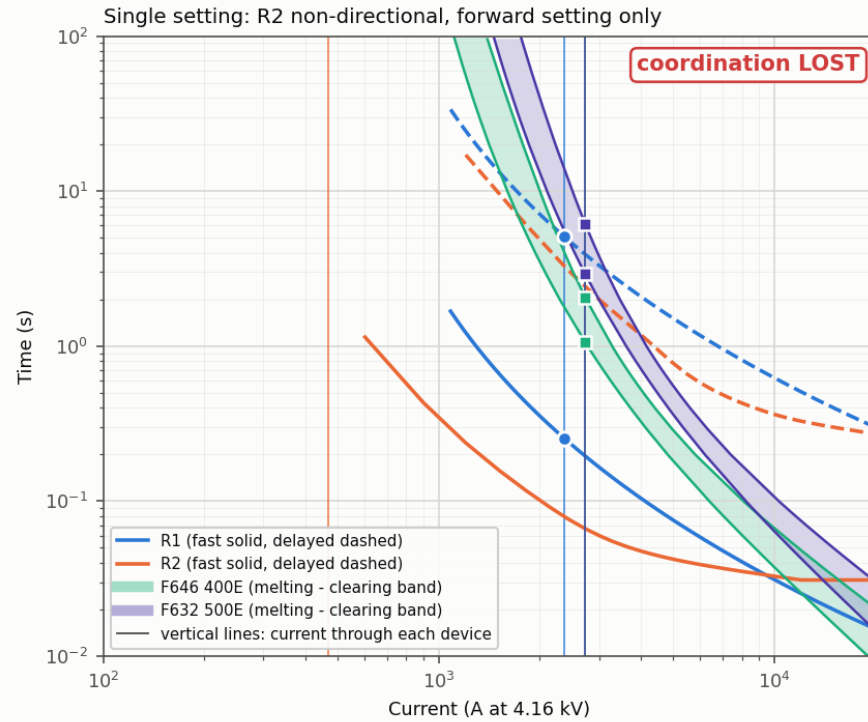
Quantity	Single setting	Dual setting (DSDR)
3-phase fault at 10 % of 632-633	LLL, phases a-b-c	(same fault and currents)
Fault impedance Z_f	0 ohm (bolted)	(same fault and currents)
Fault-path current (F633)	6398 A	(same fault and currents)
Current through R1, direction	4645 A, forward (RG60 -> 632), grid contribution	(same fault and currents)
Current through R2, direction	1777 A, REVERSE (671 -> 632), DG contribution only	(same fault and currents)
R2 setting applied (fast / delayed)	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0	REVERSE: plug 150/300 A, starts 300/600 A, TMS 0.1/1.0
R1 fast / delayed	0.084 s / 1.672 s	0.084 s / 1.672 s
R2 fast / delayed	0.121 s / 6.426 s	0.052 s / 1.470 s
Primary fuse melts / clears	F633 400E: 0.100 s / 0.163 s	F633 400E: 0.100 s / 0.163 s
Backup fuse melts	-	-
Fuse saving	LOST: F633 melts (0.100 s) before R2 fast (0.121 s)	HELD

Case 06 - LL fault at 646, $Z_f = 0$ ohm (bolted), DG 4.05 MVA connected at 692 (Figs. 15, 16)



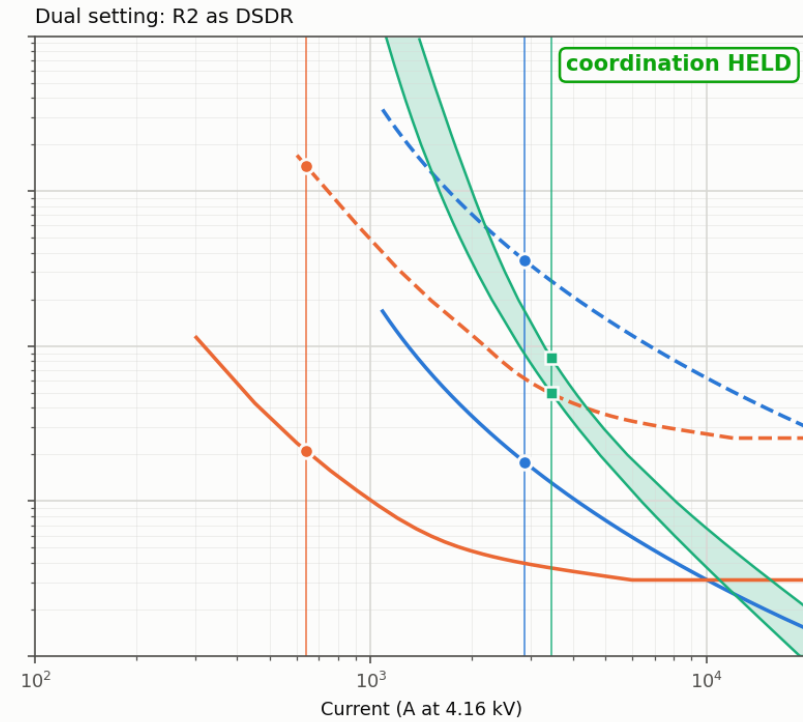
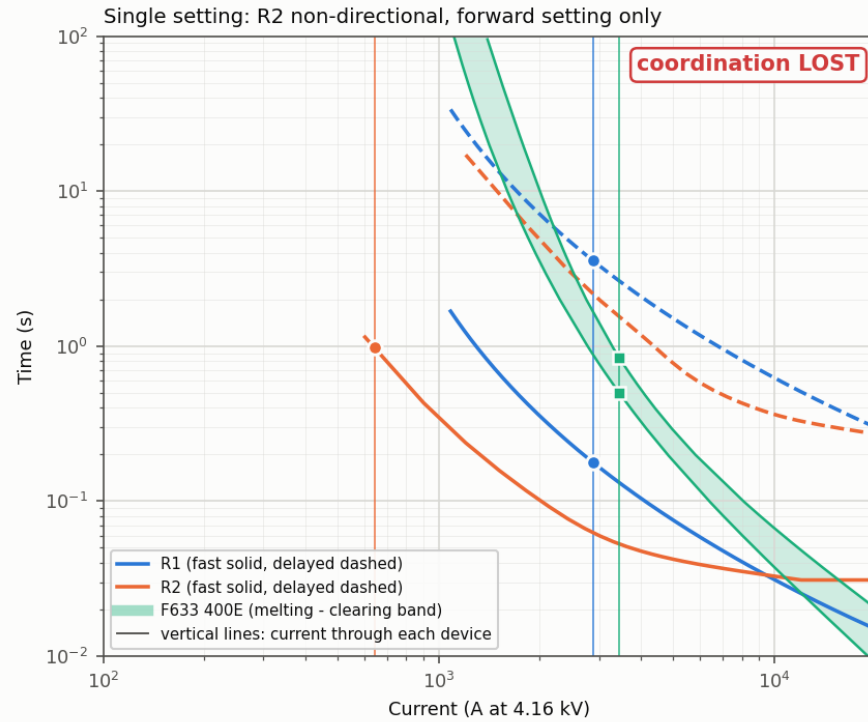
Quantity	Single setting	Dual setting (DSDR)
LL fault at 646	LL, phases BC	(same fault and currents)
Fault impedance Z_f	0 ohm (bolted)	(same fault and currents)
Fault-path current (F646)	3672 A	(same fault and currents)
Current through R1, direction	2785 A, forward (RG60 -> 632), grid contribution	(same fault and currents)
Current through R2, direction	1114 A, REVERSE (671 -> 632), DG contribution only	(same fault and currents)
R2 setting applied (fast / delayed)	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0	REVERSE: plug 150/300 A, starts 300/600 A, TMS 0.1/1.0
R1 fast / delayed	0.189 s / 3.770 s	0.189 s / 3.770 s
R2 fast / delayed	0.278 s / no trip	0.086 s / 3.771 s
Primary fuse melts / clears	F646 400E: 0.415 s / 0.684 s	F646 400E: 0.415 s / 0.684 s
Backup fuse melts	F632 500E: 0.921 s	F632 500E: 0.921 s
Fuse saving	HELD	HELD

Case 07 - LG fault at 646 (phase c), $Z_f = 0$ ohm (bolted), DG 4.05 MVA connected at 692 (Fig. 14 / 17 cell)



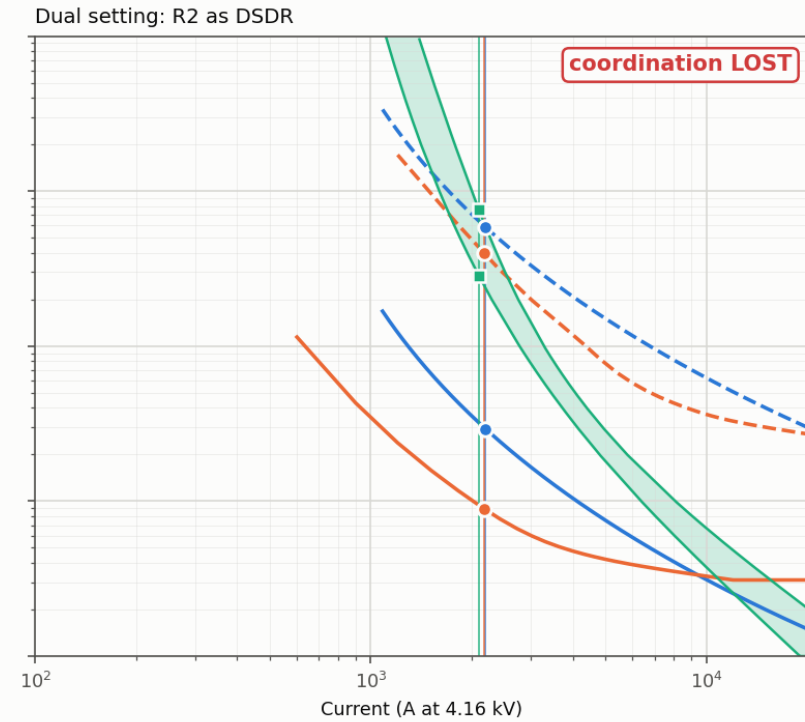
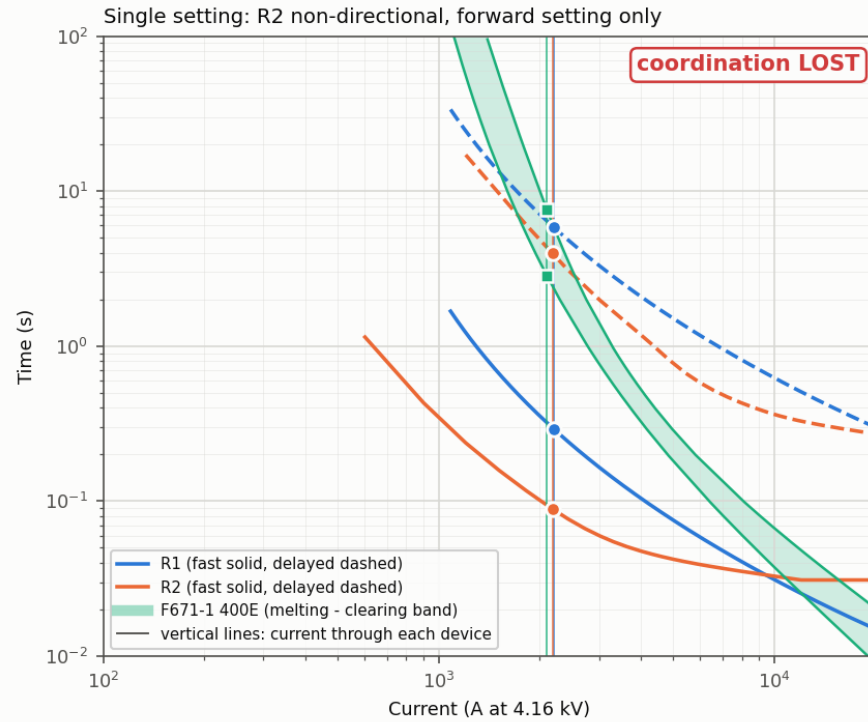
Quantity	Single setting	Dual setting (DSDR)
LG fault at 646 (phase c)	LG, phases C	(same fault and currents)
Fault impedance Z_f	0 ohm (bolted)	(same fault and currents)
Fault-path current (F646)	2732 A	(same fault and currents)
Current through R1, direction	2359 A, forward (RG60 -> 632), grid contribution	(same fault and currents)
Current through R2, direction	469 A, REVERSE (671 -> 632), DG contribution only	(same fault and currents)
R2 setting applied (fast / delayed)	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0	REVERSE: plug 150/300 A, starts 300/600 A, TMS 0.1/1.0
R1 fast / delayed	0.255 s / 5.105 s	0.255 s / 5.105 s
R2 fast / delayed	no trip / no trip	0.396 s / no trip
Primary fuse melts / clears	F646 400E: 1.057 s / 2.058 s	F646 400E: 1.057 s / 2.058 s
Backup fuse melts	F632 500E: 2.913 s	F632 500E: 2.913 s
Fuse saving	LOST: R2 does not pick up (469 A)	HELD

Case 08 - LG fault at 633 (phase b), $Z_f = 0$ ohm (bolted), DG 4.05 MVA connected at 692 (Fig. 14 / 17 cell)



Quantity	Single setting	Dual setting (DSDR)
LG fault at 633 (phase b)	LG, phases B	(same fault and currents)
Fault impedance Z_f	0 ohm (bolted)	(same fault and currents)
Fault-path current (F633)	3443 A	(same fault and currents)
Current through R1, direction	2874 A, forward (RG60 -> 632), grid contribution	(same fault and currents)
Current through R2, direction	644 A, REVERSE (671 -> 632), DG contribution only	(same fault and currents)
R2 setting applied (fast / delayed)	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0	REVERSE: plug 150/300 A, starts 300/600 A, TMS 0.1/1.0
R1 fast / delayed	0.178 s / 3.567 s	0.178 s / 3.567 s
R2 fast / delayed	0.974 s / no trip	0.210 s / 14.426 s
Primary fuse melts / clears	F633 400E: 0.502 s / 0.846 s	F633 400E: 0.502 s / 0.846 s
Backup fuse melts	-	-
Fuse saving	LOST: F633 melts (0.502 s) before R2 fast (0.974 s)	HELD

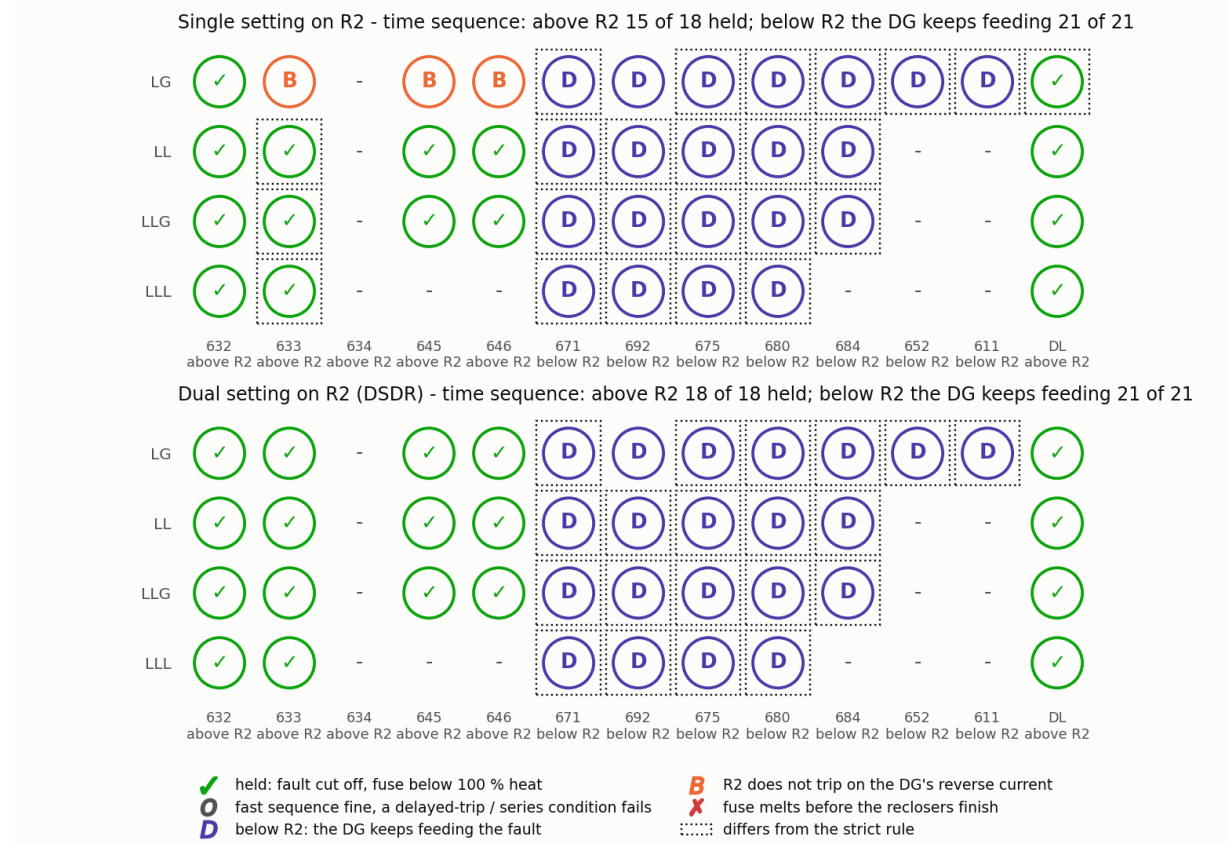
Case 09 - LG fault at 692 (phase a), $Z_f = 0$ ohm (bolted), DG 4.05 MVA connected at 692 (Fig. 14 / 17 cell)



Quantity	Single setting	Dual setting (DSDR)
LG fault at 692 (phase a)	LG, phases A	(same fault and currents)
Fault impedance Z_f	0 ohm (bolted)	(same fault and currents)
Fault-path current (F671-1)	2100 A	(same fault and currents)
Current through R1, direction	2197 A, forward (RG60 -> 632), grid contribution	(same fault and currents)
Current through R2, direction	2182 A, forward (632 -> 671), grid contribution	(same fault and currents)
R2 setting applied (fast / delayed)	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0
R1 fast / delayed	0.293 s / 5.856 s	0.293 s / 5.856 s
R2 fast / delayed	0.089 s / 3.958 s	0.089 s / 3.958 s
Primary fuse melts / clears	F671-1 400E: 2.842 s / 7.580 s	F671-1 400E: 2.842 s / 7.580 s
Backup fuse melts	-	-
Fuse saving	LOST: R2 delayed (3.958 s) before F671-1 clears (7.580 s)	LOST: R2 delayed (3.958 s) before F671-1 clears (7.580 s)

Time-sequence check of all 39 cells

The strict rule of Figs. 14 and 17 compares every recloser's fast time with the fuse's melting time at the full fault current. Here each fault is followed in time: the fuse carries the full current until the first recloser opens, then only the other source's contribution; the fuse is saved if its heat stays below 100 % of melting when the fault is cut off. The delayed-trip and series-fuse conditions are unchanged. Assumption: after one recloser opens, the other source's current stays at its contribution during the fault; nothing is simulated.



Faults above R2 (18 cells). Strict rule: 11 held with one setting, 18 with the dual setting. Time sequence: 15 and 18. Of the 7 cells the strict rule shows as restored, 3 are restored in the time sequence as well (633 LG, 645 LG, 646 LG): there the single-setting R2 does not trip on the DG's reverse current at all and the DG keeps feeding the fault. In the other 4 (633 LL/LLG/LLL, DL LG) the fuse survives with one setting too (fuse heat 32-71 %): R1 opens first and the DG's remaining current is too small to melt the fuse before R2 opens. There the dual setting shortens the time the DG feeds the fault and lowers the fuse heat.

Faults below R2 (21 cells). In all of them the DG at 692 feeds the fault directly: no recloser lies between the DG and the fault, so after R2 has opened the fault stays energised under either setting. The strict rule does not look at this. Fuse saving below R2 therefore relies on the DG's own protection disconnecting it during the dead time, which this model does not include.

Conclusion: how the DSDR improves coordination

1. The problem is the direction and size of the current through R2. For a fault above R2 (632, 633, 645, 646, DL) the fuse carries the grid's and the DG's current together (3-6 kA), but R2 carries only the DG's share, in reverse (0.5-2.0 kA). With one setting R2 measures that small reverse current against a forward setting sized for its 470 A forward load: its fast curve only starts at 600 A and its delayed curve at 1200 A.

2. With one setting R2 is blind or slow for the DG infeed. For ground faults the DG's reverse current is 470-650 A, at or below the start of the forward curve: R2 does not pick up (case 07) or needs about 1 s (case 08), so the DG keeps feeding a temporary fault and the fuse is lost.

3. The reverse setting matches the reverse duty. Its pickup follows eq. (12) from the reverse load current ($1.25 \times 252.2 \text{ A} = 315 \text{ A}$; curve from 300 A), half the forward value. The same DG currents are then cleared in 0.2-0.4 s for ground faults and in under 0.1 s for phase faults (case 06: 0.278 s $\rightarrow$ 0.086 s), before the fuse melts.

4. Nothing is given up in the forward direction. For faults below R2 (cases 01, 02, 09) both schemes use the same forward setting and give the same times; the two settings are independent.

5. In numbers. With the DG connected and the same fuses, fuse saving holds in 31 of 39 cells with one setting and in 38 of 39 with the dual setting. The 7 restored cells (633 LG/LL/LLG/LLL, 645 LG, 646 LG, DL LG) are all faults above R2, where R2 sees reverse current.

6. The limit of the claim. The dual setting only acts on R2. It cannot speed up R1, which is at its minimum dial, and it cannot help where a fuse melts faster than any recloser curve. With the fuse sizes of the no-DG design the dual setting restores only 1 cell (24 $\rightarrow$ 25 of 39); the method's fuse revision (step 9) is needed as well (24 $\rightarrow$ 31 with one setting, 38 with the dual setting). The remaining lost cell (692 LG) is a fault below R2 on the forward setting, which the DSDR does not change.

7. Faults through a fault impedance. Case 03 (LL fault at 646, $Z_f 1.0 \text{ ohm}$): R2's fast trip improves from 0.500 s to 0.131 s, but fuse saving is lost with the final fuses (series F646/F632: TCT 3.872 $>$ 0.75 $\times$ MMT 3.701) and held with the no-DG-design fuses. Case 04 (LL fault at 645, $Z_f 1.5 \text{ ohm}$): R2's fast trip improves from 0.605 s to 0.149 s, but fuse saving is lost with the final fuses (R1 delayed (10.145 s) before F632 clears (25.861 s)) and held with the no-DG-design fuses. The fuse revision was judged on bolted faults, as Figs. 14 and 17 are; at the lower currents of resistive faults the larger fuses clear slowly. This is a limit of the fuse sizes, not of the dual setting.

8. Time-sequence check. Followed in time, fuse saving above R2 holds in 15 of 18 cells with one setting and in 18 with the dual setting. The cells that are restored beyond doubt are the ground faults 633 LG, 645 LG, 646 LG, where the single-setting R2 never trips on the DG's reverse current. For the phase faults at 633 and the ground fault at DL the strict rule overstates the loss: the fuse would survive with one setting too, and the dual setting adds margin. Below R2 the DG feeds every fault directly, whatever R2's setting.

Conclusion. The DSDR improves coordination because it lets R2 detect and interrupt the DG's reverse fault contribution with a setting sized for that contribution, instead of a forward setting sized for the grid. That restores the fuse-saving scheme for every fault above R2 without touching the forward coordination. It is a necessary part of the solution in this feeder, but not sufficient alone: it works together with the fuse revision.